perfect information A game of perfect information is a game in which every player can observe the complete current state of the game.
Deterministic Games each action always leads to a predictable and fixed next state.
Zero-Sum Games one agent's gain is another agent's loss.

- Agents have opposing utilities (values of outcomes).
- If one agent wants to maximize its utility, the other agents want to minimize it.
- Adversarial and purely competitive.

General Games each agent has its own independent utility function.

- Each agent has its own independent utility function.
- Agents may cooperate, ignore each other, compete, or interact in various ways.

Adversarial search is a search method used in competitive games. The agent tries to maximize its utility while the opponent tries to minimize it. (Tic-Tac-Toe \Chess \Checkers)
Minimax Search is an adversarial search algorithm that evaluates a game tree by letting MAX choose the maximum value and MIN choose the minimum value.

- Uses a state-space search tree.
- Players take turns acting.
- Computes the minimax value of each node.

The minimax value represents: The best outcome achievable against a rational and optimal opponent.
①**Completeness** Assuming the game tree is finite. If a solution exists, the algorithm will find it.
②**Optimality** Assuming the opponent also plays optimally. The algorithm finds the optimal solution.
③**Time Complexity** $O(b^m)$
where: b= branching factor m= maximum depth of the search tree
④**Space Complexity** $O(bm)$

Resource Limitations
Problem In realistic games, it is impossible to search all the way to the leaf nodes.
Solution Depth-limited search. Limit the depth of the search tree. Replace terminal utilities with an evaluation function at non-terminal nodes. Utility→Eval
Evaluation Functions In depth-limited search, evaluation functions assign scores to non-terminal states.
Ideally: An evaluation function returns the true minimax value of a state.

In practice: It is usually a weighted linear combination of features. $\boxed{Eval(s) = w_1 f_1 + w_2 f_2 + \cdots + w_n f_n}$

Example:$f_1(s)$ = (number of White queens) − (number of Black queens)
Horizon Effect In depth-limited search, the algorithm cannot see beyond the cutoff depth (the horizon). When a state appears to have a high evaluation value, but a much worse state exists just beyond the search horizon.
Thrashing Thrashing is a phenomenon in which an agent repeatedly switches between plans or actions without making progress toward its goal.

Alpha-Beta Pruning

1. Search the game tree like Minimax.像 Minimax 一样搜索游戏树。
2. MAX chooses the largest value; MIN chooses the smallest value. MAX 选最大值， MIN 选最小值。
3. Keep updating α for MAX and β for MIN.
4. If (α⩾β), prune the rest.剪掉剩余分支。

$$O(b^{m/2})$$

Monte Carlo Tree Search (MCTS) MCTS repeatedly selects a promising node, expands a new child, performs a rollout(play the games randomly) to the end of the game, and propagates the result back through the tree. In **AlphaGo**, reinforcement learning was used to identify more promising moves, quickly evaluate them, and focus the search on the most promising regions of the tree.